
Verification Before Generation Standard

Standard: CQD-515D

Edition: 1

Status: draft for comment

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Abstract.

This standard template defines verification before generation as the order of operations for embodied clinical code, as numbered clauses with conformance requirements.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 1 Scope

This standard specifies the requirement that an automated verification, validation, and uncertainty quantification process clear robot-patient interaction code before that code is generated or executed.

2 2 Normative references

The following referenced documents are indispensable for the application of this standard.

- ASME V&V 40-2018 credibility of computational models
- ISO/TS 15066 collaborative robot contact limits
- IEC 80601-2-77 safety of robotically assisted surgical equipment

3 3 Terms and definitions

For the purposes of this standard, the following terms apply.

- robot-patient interaction code software governing a robot's contact behavior
- gate an automated check emitting ACCEPT, BLOCK, or ESCALATE
- verification fraction the share of verification checks that pass

4 4 Requirements

Conformance requirements use shall.

- 4.1 code shall not be generated or executed unless the gate has first cleared
- 4.2 the verification fraction shall equal 1.0 for each interaction
- 4.3 three catastrophe gates shall hold agreement 1.00

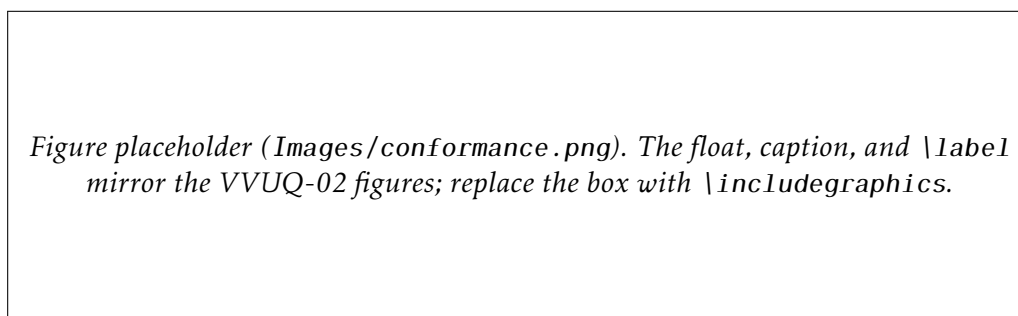


Figure 1: Figure 1. Placeholder schematic of the conformance flow.

5 5 Conformance

Clause	Requirement	Conformance
4.1	verify before generate	record timing
4.2	fraction 1.0	exact proportion
4.3	catastrophe 1.00	hard predicate

Table 1: Table 1. Conformance summary (modeled on VVUQ-05 Table 1).

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Cite This Standard

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Data Availability

Supporting records are openly available in the [kevinkawchak/cancer-automated](#) and [kevinkawchak/physical-ai-oncology-trials](#) GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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